BRAINSTORM GROUP (BSG)

MAT111
Elementary Mathematics I
Compiled Practice Questions

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SET THEORY

1. Given that $M = \{x: x < 4\}$ and $N = \{x: x \geq -2\}$ where x is an integer, find $M \cap$ N
   A. $\{0, 1, 2, 3\}$ B. $\{0, 1, 2, 3\}$ C. $\{-2, -1, 1, 2, 3\}$ D. $\{-2, -1, 0, 1, 2, 3\}$

2. Given $P = \{x: x\ is\ a\ factor\ of\ 24\}$, then $n(P) =$
   A. 4 B. 6 C. 8 D. 12 E. 24

3. Let $U = \{1, 2, 3, 4, \ldots, 10\}$, $A = \{odd\ numbers\ up\ to\ 9\}$ and $B = \{numbers\ less\ than\ 7\}$, find $(A \cap$ B)
   A. $\{3, 7\}$ B. $\{1, 3, 7\}$ C. $\{1, 3, 5\}$ D. $\{3, 5\}$

4. Given $Q = \{x: x + 3 \leq 7, x\ is\ an\ integer\}$, then $n(Q)$
   A. 5 B. 4 C. 6 D. ∞

5. Let $U = \{1, 2, 3, 4, \ldots, 10\}$, $A = \{odd\ numbers\ up\ to\ 9\}$ and $B = \{numbers\ less\ than\ 7\}$, find $(A \cup$ B$)'$
   A. $\{0, 10\}$ B. $\{1, 10\}$ C. $\{8. 10\}$ D. $\{9, 10\}$

6. In a village all the people speak Hausa or English or both. If 97% speak Hausa and 64% speak English, what percentage speak both languages?
   A. 61% B. 63% C. 67% D. 69%

7. In a school of 750 students, 320 are girls. 559 students do some kind of sport. If 101 girls do no sport, how many boys also do no sport?
   A. 80 B. 90 C. 100 D. 110

8. Given $P = \{x: x\ is\ a\ factor\ of\ 24\}$, which of the following is not a subset of P?
   A. $\{1, 2, 3, 4\}$ B. $\{2, 4, 6, 8\}$ C. $\{2, 4, 8, 16\}$ D. $\{3, 6, 12, 24\}$

9. Given $X = \{x: x\ is\ a\ factor\ of\ 24\}$ and $Y = \{x: x + 3 \leq 7, x\ is\ not\ negative\}$ find $X \cap$ Y
   A. $\{1, 2, 3\}$ B. $\{1, 2, 4\}$ C. $\{1, 2, 3, 4\}$ D. $\{0\ 1, 2, 3, 4\}$

10. Out of 30 students, 21 do sports and 11 are girls. If 10 of the boys do sports, how many of the girls do not do sports?
    A. 9 B. 10 C. 11 D. 0

11. If X = $\{1, 2, 3, 4\}$ and Y = $\{3, 5, 6\}$, the elements of $(X \cap Y) \cup$ X are
    A. $\{3\}$ B. $\{1, 2, 3, 4\}$ C. $\{1, 2, 5, 6\}$ D. $\{1, 2, 3, 4, 5, 6\}$

12. If $\mathcal{U}$ = $\{5, 6, 7, 8\}$, P = $\{6, 7\}$ and Q = $\{6, 8\}$ then $(P \cap Q)'$ is
    A. $\{6\}$ B. $\{7, 8\}$ C. $\{6, 7, 8\}$ D. $\{5, 7, 8\}$

13. In a village of 650 youths, 250 are girls, 425 of the youths like music. If 90 girls do not like music, how many boys do not like music?
A. 135 B. 160 C. 175 D. 310 E. 335

14. In a class of 20 boys, 16 play soccer, 12 play hockey and 2 are not allowed to play games. Find how many students play soccer and hockey.
A. 6 B. 8 C. 10 D. 12

15. $P - Q$ is same as
A. $(P \cap Q)$ B. $(P \cap Q)'$ C. $(P \cap Q')$ D. $(P' \cap Q)$

16. Given $n(A - B) = 18, n(A \cup \mathrm{B}) = 70\ and\ n(A \cap \mathrm{B}) = 25.\ Find\ n(B)$.
A. 27 B. 43 C. 45 D. 52

17. In a market place all the people buy or sell things or do both. If 70% of the people do buy and 90% do sell, what percentage of people do both?
A. 60% B. 20% C. 50% D. 30%

18. Given that $P = \{1,\ 3,\ 5,\ 7\}\ and\ Q = \{2,\ 5,\ 6,\ 8,\ 9\}.\ Find\ (P \cup Q)$.
A. {1, 2, 3, 5, 6, 7, 8, 9} B. {1, 2, 3, 6, 7, 8, 9}
C. {1, 2, 3, 4, 5, 6, 7, 8, 9} D. {3, 5}

19. In a class of 45 students, 25 take Economics and 21 take Home Management. If 3 take neither of the two, how many take Economics only?
A. 4 B. 17 C. 21 D. 24

20. Which of the following is not a subset of P if P = {2, 4, 6}.
A. {2, 2, 4} B. {2, 3, 4} C. {2, 4, 6} D. ∅

21. $(M' \cap N)'$ is equivalent to
A. $(M \cap N')'$ B. $(M' \cap N)$ C. $(M \cup N')$ D. $(M \cup N)$

22. In a school of 65 teachers, 45 teachers teach pure Science, 32 teach Social Science. If 7 teachers do not teach any subject, how many teach both?
A. 5 B. 12 C. 13 D. 19

23. There are 50 staff in a Specialist Hospital, 35 are qualified nurses and 20 are dentists. If 4 are neither, how many are both nurses and dentists?
A. 4 B. 5 C. 9 D. 11

24. If $P \subset Q, then\ (P \cup Q) \cap P\ is$ is
A. ∅ B. P C. Q D. $\mathcal{U}$

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25. The cardinality of the set Z = {M, I, S, S, I, S, S, I, P, P, I} is

A. 4 B. 6 C. 8 D. 11

26. $(A \cap A')'$ is equivalent to

A. $\emptyset$ B. A C. A' D. $\mathcal{U}$

27. Given that U = {x : x is an integer, $2 \leq x \leq 10$}, $A = \{x : x \text{ is a prime number}\}, B = \{x: \text{ is a multiple of } 3\}$. *Find* $n(A \cup B)$.

A. 1 B. 3 C. 4 D. 6

28. Given that $X = \{x : 3 < x < 6\}$ *and* $Y = \{x : 4 \leq x \leq 8\}$. *Find* $X \cap Y$.

A. {4} B. {4, 5} C. {5, 6} D. {4, 5, 6, 7, 8}

29. In a certain school 21% of the teachers were trained in Britain, 26% were trained in U.S.A. If 15% of the teachers were trained in both countries, what is the percentage of teachers who were not trained in either country?

A. 32% B. 62% C. 38% D. 68%

30. If set $F = \{2 < x \leq 5\} \cup \{9 < x < 12\}$. List the members of F.

A. $\emptyset$ B. {2, 3, 4, 5, 9, 10, 11, 12}
C. {3, 4, 5, 6, 9, 10, 11} D. {3, 4, 5, 10, 11}

31. In a class of 20 boys, 16 play soccer, 12 play hockey and 2 are not allowed to play games. Find how many students play only hockey.

A. 2 B. 6 C. 10 D. 12

32. Let $U = \{1, 2, 3, 4, \ldots, 10\}, A = \{\text{odd numbers up to } 9\}$ and $B = \{\text{numbers less than } 7\}$, find $n(A \cup B)$

A. 7 B. 8 C. 9 D. 10

33. All inhabitants of a certain country speak either French and/or English. If 60% speak French and 50% speak English, what percentage of them speak both languages?

A. 10% B. 30% C. 70% D. 110%

34. $(A \cup A')$ is equivalent to

A. $\emptyset$ B. A C. A' D. $\mathcal{U}$

35. Given that $X = \{x : 3 < x < 6\}$ *and* $Y = \{x : 4 \leq x \leq 8\}$. *Find* $X \cap Y$.

A. {4} B. {4, 5} C. {5, 6} D. {4, 5, 6, 7, 8}

36. If Q = {all perfect squares less than 30} and P = {all odd numbers from 1 to 10}. Find $(Q \cap P)$.

A. {0, 1, 4, 9} B. {1, 4, 9} C. {1, 4} D. {4, 9} E. {1, 9}

37. Given that P and Q are set of real numbers such that $P = \{x : 0 \leq x \leq 3\}$, $Q = \{x : -3 < x < 2\}$, find $P \cup Q$

A. $\{-3 \leq x < 3\}$ B. $\{-3 < x < 3\}$ C. $\{-3 < x \leq 3\}$ D. $\{-3 \leq x \leq 2\}$

38. $P = \{3, 9, 11, 13\}$ and $Q = \{3, 7, 9, 15\}$ are subsets of universal set $U = \{1, 3, 7, 9, 11, 13, 15\}$. Find $P' \cap Q'$

A. $\{3, 9\}$ B. $\{3\}$ C. $\{1\}$ D. $\{3, 7, 9\}$

39. The sets $P = \{2, 5, 8\}$ and $Q = \{2, 3, 5, 7\}$ are subsets of the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find $P' \cap Q$

A. $\{3\}$ B. $\{3, 5\}$ C. $\{3, 7\}$ D. $\{1, 3\}$

40. The sets $P = \{2, 5, 8\}$ and $Q = \{2, 3, 5, 7\}$ are subsets of the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find $n(P \cap Q')$

A. 1 B. 2 C. 3 D. 0

41. A die is thrown once, what is the value of all possible subsets on the die?

A. 60 B. 62 C. 64 D. 68

42. List the element of the set $Y = \{x: x^2 - 9 = 0; x > 0\}$

A. $\{-3, 3\}$ B. $\{\}$ C. $\{3\}$ D. $\{-3\}$

43. List $n(even\ numbers)$

A. 100 B. 200 C. infinite D. 400

44. Find the cardinality of set $P = \{x: 3 < x \leq 20\}$

A. 16 B. 17 C. 18 D. 19

45. Find the cardinality of $n \in \mathbb{N} : 3n > 12$

A. 4 B. 3 C. infinity D. $\{0, 1, 2, 3, 4\}$

46. What is the value of $n(\{3, 5\})$

A. 1 B. 2 C. 3 D. 4

47. List the element of the set $X = \{n^2 + 1;\ n = 0, 1, 2, 3\}$

A. $\{0, 2, 4, 9\}$ B. $\{1, 2, 5, 10\}$ C. $\{0, 1, 2, 3\}$ D. $\{0, 2, 5, 10\}$

48. Find $n(\{1, \{2, 3\}, \{4, 5, 6\}\})$

A. 1 B. 2 C. 3 D. 6

49. Given that $S = \{All\ states\ in\ Nigeria\}$, what is the cardinality of S?

A. 34 B. 36 C. 37 D. 38

50. Which of the following set description is not correct?

A. $B = set\ of\ Geometry\ book$ B. $A\ \{x; even\}$ C. $H\ \{1, 2, 3: 4\}$ D. $S = \{a, b, c\}$

51. What is the value of $n(\{\})$

A.* 0 B. 1 C. 2 D. 3

52. What is the value of $n(\{a, a\})$

A. 0 B.* 1 C. 2 D. 3

53. In an examination, 18 students passed MAT111, 17 students passed PHY113, 11 students passed both subject. Find the number of students that passed MAT111 only.

A. 5 B. 6 C.* 7 D. 8

54. If $F = \{x: (2 < x \leq 5) \cup (9 < x < 12)\}$, list the elements of set F

A. $\{2, 3, 4, 5, 9, 10, 11, 12\}$ B. $\{3, 4, 5, 9, 10, 11\}$ C.* $\{3, 4, 5, 10, 11\}$ D. $\{3, 4, 5, 10, 11, 12\}$

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MAPPING AND FUNCTION

1. If $f: x = 1 + 2x$ and $g: x = \frac{1}{1+x}$, find $fg\left(\frac{1}{2}\right)$

   A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{5}{3}$ D. $\frac{7}{3}$

2. If $f: x = 1 + 2x$ and $g: x = \frac{1}{1+x}$, find $f^{-1}g^{-1}\left(\frac{2}{3}\right)$

   A. $\frac{1}{4}$ B. $-\frac{1}{2}$ C. $-\frac{1}{4}$ D. $-\frac{3}{4}$

3. Given the function $f(x) = 5x - 2$, find $f^{-1}(-1)$

   A. $-\frac{1}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{3}{5}$

4. Given the mapping $f: x = 7x + 2$, determine $f(-\frac{1}{2})$

   A. $-\frac{5}{2}$ B. $-\frac{3}{2}$ C. $\frac{1}{2}$ D. $\frac{5}{2}$

5. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute $(fog)(2)$

   A. 25 B. 49 C. 98 D. 89

6. If $f: x \to \frac{2}{3(2-7x)}, x \neq \frac{2}{7}$, find $f^{-1}(x)$

   A. $\frac{6x-2}{21x}$ B. $\frac{2-6x}{21x}$ C. $\frac{6x-2}{7x}$ D. $\frac{2-6x}{7x}$

7. Two functions f and g are defined on the set R of real numbers by: $f(x) = x^2 - 2x - 1$ and $g(x) = x - 1$. Find the value of x for which $f(x) = g(x) - 2$

   A. 0 *or* 1 B. 1 *or* 2 C. 0 *or* 2 D. 1 *or* 3

8. Two functions f and g are defined on the set R of real numbers as follows: $f: x = 2x - 3$ and $g: x = x^2 + 5$. Find $f[g(x-1)]$

   A. $2x^2 - 4x + 12$ B. $2x^2 - 4x + 9$ C. $x^2 - 2x + 6$ D. $2x^2 - 2x + 12$

9. Given that image of p under the mapping $x \to 3x + 2$ is -10, what is the value of p?

   A. -4 B. -3 C. -2 D. -1

10. Let $f: x = x^2 + 1, x\ e\ R$ find $f(1) - f(0)$

    A. 2 B. -1 C. 1 D. -2

11. If $f(x) = 2(x-3)^2 + 3(x-3) + 4$ and $g(y) = 5 + y$. Find $f[g(-1)]$

    A. 2 B. 3 C. 4 D. 9

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12. $h: x = 3x - 2$ is a mapping defined on the set R of real numbers. Determine the pre-images of $\{-5, 1, 7\}$.

A. $\{-17, 1, 19\}$ B. $\{-1, 1, 3\}$

C. $\{-1, 1, 2\}$ D. $\{-3, 1, 3\}$

13. Given the function $h(x) \rightarrow \frac{1}{2}e^{2-x} - 2$, find the inverse function.

A. $2 + ln[2(x-2)]$ B. $ln[2(x+2)] - 2$

C. $2 - ln[2(x-2)]$ D. $2 - ln[2(x+2)]$

14. If $g: x \rightarrow \frac{x+3}{3x-2}, x \neq \frac{2}{3}$, find the value for which $g^{-1}(x)$ is zero.

A. $-\frac{3}{2}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. $-\frac{2}{3}$

15. If every element in the domain has different image in the co-domain, such type of mapping is called

A. constant mapping B. onto mapping

C. one-one mapping D. composite mapping

16. A mapping f is called _____ if every element of the co-domain is an image of at least one element in the domain?

A. constant mapping B. onto mapping

C. one-one mapping D. composite mapping

17. The rule that assigns an element to two or non-empty set is

A. logic B. set C. mapping D. rules

18. Given $f(y) = py$ and $f(3) = 81$, determine the value of p.

A. 3 B. 9 C. 18 D. 27

19. If $g: x \rightarrow \frac{x+3}{3x-2}, x \neq \frac{2}{3}$, find the value for which $g^{-1}(x)$ is zero.

A. $-\frac{3}{2}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. $-\frac{2}{3}$

20. Given the inverse of a function to be $f(x) = \sqrt{4ax + b}$, find the main function

A. $\frac{x^2+b}{4a}$ B. $\left(\frac{x-b}{4a}\right)^2$ C. $\frac{x^2-b}{4a}$ D. $\left(\frac{x+b}{4a}\right)^2$

21. Find the inverse of $f: x = \frac{x+1}{x-2}$.

A. $\frac{2x+1}{x-1}$ B. $\frac{2x+1}{x+1}$ C. $\frac{2x-1}{x-1}$ D. $\frac{2x-1}{x+1}$

22. Determine the zero of the function defined by $h(x) = x^2 + x - 6$.

A. -2 B. 3 C. 1 D. 2

23. If $f(g(x)) = x^2 + 2x + 1$ and $f(x) = x^2$, find $g(x)$.

A. $x + 2$ B. $x - 2$ C. $x + 1$ D. $x^2 + 2$

24. The function of f and g are defined over R as $f: x = x^2 + 2$ and $g: x = \frac{2x}{1-x}$. Find $fog(3)$.

A. -3 B. -9 C. 9 D. 11

25. Let $f: x = \frac{x-1}{x-2}$ where $x \neq 1, x \neq 2$, find $f^{-1}(x)$.

A. $\frac{2x-1}{x-1}$ B. $\frac{2x+1}{x-1}$ C. $\frac{2x-1}{x+1}$ D. $\frac{2x+1}{x+1}$

26. Determine the zeros of the function defined by $h(x) = x^2 - x$

A. $0\ and\ 1$ B. $0\ and\ -1$ C. $1\ and\ -1$ D. $0\ and\ 2$

27. Given that $f(x) = x^5 + 4x^4 - 6x^2 + 2x + 2$, find $f(-1)$.

A. -3 B. -2 C. -1 D. 1

28. What is the set of values of x for which the mappings $x \to x^2$ and $x \to x^3$ have the same functional values?

A. $\{-1, 0, 1\}$ B. $\{1\}$ C. $\{0, 1\}$ D. $\{2\}$

29. A relation $g: R \to R$ is defined by $g(x) = \frac{2}{x-2}, x \neq 2$. Where R is the set of real numbers, find $g^{-1}[g(x)]$.

A. x B. $2x$ C. 0 D. 1

30. Determine the domain D of the mapping $g: x \to 2x^2 - 1$, if $R = \{1, 7, 17\}$ is the range and g is defined on D.

A. $\{1, 2, 3\}$ B. $\{0, 1, 2, 3\}$ C. $\{-3, -2, -1, 0, 1, 2, 3\}$ D. $\{-3, -2, -1, 1, 2, 3\}$

31. If $f(x) = 2(x-3)^2 + 3(x-3) + 4$ and $g(y) = 5 + y$. Find $g[f(3)]$.

A. 4 B. 5 C. 9 D. 0

32. Given the function $h(x) \to e^{x-1} + 3$, find the inverse function.

A. $ln(x-3) + 1$ B. $ln(x+3) + 1$ C. $ln(x-3) - 1$ D. $ln(x+3) - 1$

33. Find the inverse function of $g(a) \to 2e^{2a} - 2$.

A. $ln\left(\frac{a+2}{2}\right)$ B. $2ln\left(\frac{a+2}{2}\right)$ C. $\frac{1}{2}ln(a+2)$ D. $\frac{1}{2}ln\left(\frac{a+2}{2}\right)$

34. Let $f: x = x^2 + 1, x\ R$ find $f(-2) - f(-3)$.

A. -3 B. 3 C. -5 D. 5

35. Find the range of $f: x \to x^2 - 1$, if the domain is $\{-2, -1, 0, 1, 2\}$.

A. $\{-1, 0, 3\}$ B. $\{-1, 0, 1, 2, 3\}$ C. $\{1, 0, 3\}$ D. $\{-1, 0, 2\}$

36. Find the image of $\left(\frac{1}{3},\frac{2}{3}\right)$, under the mapping $(x,y) \rightarrow \left(\frac{x+y}{y},\ y\right)$.

A. $\left(-\frac{2}{3},\frac{2}{3}\right)$ B. $\left(-\frac{3}{2},\frac{2}{3}\right)$ C. $\left(\frac{3}{2},\frac{2}{3}\right)$ D. $\left(\frac{2}{3},\frac{3}{2}\right)$

37. If $f: x = 2x - 3$ and $g: x = 3x + p$, for what value of constant p is $fog = gof$.

A. -3 B. 3 C. -2 D. -6

38. If $g: x \rightarrow \frac{x+3}{3x-2}, x \neq \frac{2}{3}$, find the value for which $g^{-1}(x)$ is not defined.

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $1\frac{1}{3}$ D. $2\frac{1}{3}$

39. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute (fog).

A. $9x^2 + 6x + 1$ B. $18x^2 + 12x + 1$
C. $18x^2 + 12x + 2$ D. $18x^2 + 12x - 2$

40. Given that $f: x = ax^2 + bx + 1$ is a mapping on the set of real numbers. Given that $f(1) = 5$ and $f(-1) = 3$, determine the values a and b.

A. $a = 3, b = 2$ B. $a = 2, b = 3$
C. $a = 3, b = 1$ D. $a = 1, b = 3$

41. Determine the zero of the function $2x - 3$

A. $-\frac{3}{2}$ B. -3 C. $\frac{3}{2}$ D. $-\frac{2}{3}$

42. Find the range of $y = 7x + 2$

A. $-\infty, 0$ B. $0, \infty$ C. $-\infty, \infty$ D. $0, 1$

43. The function f is given as $f(x) = ln(4x - 2)$, find in $f^{-1}(x)$ its simplest form

A. $\frac{1}{4}(e^x - 2)$ B. $-\frac{1}{4}(e^x - 2)$ C. $\frac{1}{4}(e^x + 2)$ D. $-\frac{1}{4}(e^x + 2)$

44. The function f is given as $f(x) = 3 - lnx$, find in $f^{-1}(x)$ its simplest form

A. $\frac{1}{3}e^x$ B. e^{x-3} C. e^{3-x} D. $3e^x$

45. The functions f and g are given by $f(x) = x^2$ and $g(x) = \frac{1}{x+2}$, solve the equation $fg(x) = \frac{4}{9}$

A. $\frac{1}{2}$ *and* $\frac{7}{2}$ B. $-\frac{1}{2}$ *and* $\frac{7}{2}$ C. $\frac{1}{2}$ *and* $-\frac{7}{2}$ D. $-\frac{1}{2}$ *and* $-\frac{7}{2}$

46. The functions f and g are given by $f(x) = ln(4 - 2x)$ and $g(x) = e^{3x}$, solve the equation $fg(x) = 0$

A. $\frac{1}{2}ln\frac{3}{2}$ B. $\frac{3}{2}ln\frac{1}{2}$ C. $2ln\frac{3}{2}$ D. $-\frac{1}{2}ln\frac{3}{2}$

47. The functions f and g are given by $f(x) = (x-3)^2 + 1, x \in \mathbb{R}, x \geq 4$, solve the equation $f(x) = 17$

A. -1 B. $+1$ C. -7 D. $+7$

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48. The functions g and f are given by $g(x) = 4 - 3x$ and $f(x) = x^2 + ax + b$, where a and b are non-zero constants. Given that $fg(2) = -5$ and $gf(2) = -29$, find the value of a and b.

A. $a = -1, b = 4$ B. $a = 1, b = -4$

C. $a = 4, b = 1$ D. $a = 4, b = -1$

49. The function f is given as $f(x) = \frac{x+6}{x+2}, x \in \mathbb{R}, x \neq -2$, solve the equation $f(x) = f^{-1}(x)$

A. $-3, -2$ B. $3, -2$ C. $-3, 2$ D. $3, 2$

50. The function h is given as $h(x) = \frac{x}{x-1}, x \in \mathbb{R}, x \neq 1$, find in its simplest form the composition $hh(x)$

A. $\frac{x-1}{x}$ B. $x - 1$ C. $1 - x$ D. x

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COMPLEX NUMBER

1. *Given* $Z = 4 - 2\sqrt{2}i$, *find* $Z\bar{Z}$

   A. 20 B. 22 C. 24 D. 21

2. *Find the imaginary part of* $\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^2$

   A. $-\frac{1}{2}\sqrt{3}i$ B. $\frac{1}{2}\sqrt{3}i$ C. $-\frac{1}{4}\sqrt{3}i$ D. $\frac{1}{4}\sqrt{3}i$

3. *Find* $\frac{Z_1}{Z_2}$ *if* $Z_1 = 2 - 4i$ *and* $Z_2 = 6 + 7i$

   A. $\left(\frac{16}{85} + i\frac{38}{85}\right)$ B. $\left(\frac{15}{85} - i\frac{38}{85}\right)$ C. $-\left(\frac{16}{85} - i\frac{38}{85}\right)$ D. $-\left(\frac{16}{85} + i\frac{38}{85}\right)$

4. *If* $Z_1 = 1 - 3i$ *and* $Z_2 = -2 + 5i$, *determine* $\frac{Z_1 Z_2}{Z_1 + Z_2}$ *in the form* $a + ib$

   A. $-\frac{9}{5} + \frac{37}{5}i$ B. $\frac{9}{5} - \frac{37}{5}i$ C. $-\frac{12}{25} - \frac{13}{25}i$ D. $\frac{9}{25} - \frac{37}{25}i$

5. *Find the real part of* $\frac{2+3i}{3+2i}$

   A. $\frac{12}{13}$ B. $-\frac{5}{13}$ C. $\frac{5}{13}$ D. $-\frac{12}{13}$

6. *What is the value of a given* $a + ib = \frac{1-i}{2+i}$

   A. $-\frac{1}{5}$ B. $\frac{3}{5}$ C. $-\frac{3}{5}$ D. $\frac{1}{5}$

7. Given that $Z_1 = 2 + 3j$ and $Z_2 = 3 - 4j$, evaluate $Z_2 - Z_1$

   A. $1 + 7j$ B. $1 - 7j$ C. $-1 + 7j$ D. $-1 - 7j$

8. Given that $Z_1 = 2 + 3j$ and $Z_2 = 3 - 4j$, evaluate $Z_1 + Z_2$

   A. $1 - 5j$ B. $5 - j$ C. $6 - 12j$ D. $12 - 6j$

9. *Given* $Z_1 = (3 + 2i)$ *and* $Z_2 = (4 - 5i)$, *evaluate* $Z_1 Z_2$

   A. $2 - 7i$ B. $22 - 7i$ C. $2 + 7i$ D. $22 + 7i$

10. Given $2 - 3i = \sqrt{a + bi}$. Find a and b.

    A. -5 and -12 B. 5 and -12 C. -5 and 12 D. 5 and 12

11. If $(x - 2yi) - (y - xi) = 2 + i$, find x.

    A. -3 B. -1 C. 3 D. 1

12. Given $A = 1 + 2i$, the conjugate of A is

    A. $-1 + 2i$ B. $-1 - 2i$ C. $1 - 2i$ D. $1 + 2i$

BRAINSTORM GROUP

13. Determine the modulus, argument of $2 + 4i$.

A. $2\sqrt{5},\ 63.43°$ B. $2\sqrt{10},\ 63.43°$ C. $2\sqrt{5},\ 26.57°$ D. $2\sqrt{10},\ 26.57°$

14. Simplify $i^9 + 2i^{11} + i^{13}$.

A. -4 B. -1 C. 1 D. 0

15. Evaluate $\frac{1-j}{1+j}$

A. $-j$ B. j C. 1 D. -1

16. Find $argZ\ if\ Z = i - 1$.

A. $\frac{1}{2}\pi$ B. $\frac{2}{3}\pi$ C. $\frac{4}{3}\pi$ D. $\frac{3}{4}\pi$

17. Given $(1 + 2i)(-2 - 3i) = a + bi$. Find $a + b$.

A. -3 B. 3 C. -4 D. 4

18. Simplify $(i - 1)^4$.

A. $-4i$ B. -4 C. $4i$ D. 4

19. The product of a complex numbers and its conjugate results in a _____ number.

A. real B. imaginary C. complex D. natural

20. If $Z_1 = 1 + 2i\ and\ Z_2 = 4 - 3i$, what is the value of $|Z_2 - Z_1|$?

A. $\sqrt{34}$ B. $\sqrt{16}$ C. $2\sqrt{5}$ D. 4

21. Evaluate $-\frac{4}{i^9}$

A. -4 B. 4 C. $-4i$ D. $4i$

22. Evaluate $-\frac{1}{i^7}$

A. $-i$ B. i C. 1 D. -1

23. Evaluate $\frac{2}{(1+i)^4}$

A. -2 B. 2 C. $-\frac{1}{2}$ D. $\frac{1}{2}$

24. _____ numbers ranges from $-\infty$ to $+\infty$

A. real B. integer C. complex D. natural

25. Find the solution of the equation $x^2 + 2x = -5$

A. $-1 \pm 2i$ B. $1 \pm 2i$ C. $-2 \pm i$ D. $2 \pm i$

26. Simplify $i^3(1 + i) + i^5(3 - i) + i^7(2 + i)$

A. 0 B. 2 C. 3 D. 4

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27. If $\frac{2+j}{1-j} = j(x + yj)$. Find y and x.

A. $\frac{1}{2}, \frac{3}{2}$ B. $\frac{3}{2}, \frac{1}{2}$ C. $\frac{1}{2}, -\frac{3}{2}$ D. $-\frac{1}{2}, \frac{3}{2}$

28. Determine the modulus, argument of $-5 - 2i$.

A. $5.385, -158.2°$ B. $5.385,\ 26.57°$ C. $4.583, -158.2°$ D. $4.583,\ 26.57°$

29. If $z = 2 + i$ and $w = 3 - i$, evaluate $3z - 2w$

A. $5i$ B. $-5i$ C. 5 D. -5

30. If $w = 3 - i$ and $z = 2 + i$, evaluate $wz^2 - 1$

A. $-12 + 9i$ B. $12 - 9i$ C. $12 + 9i$ D. $-12 - 9i$

31. The complex conjugate of $3 + 4i$ is

A. $-4i + 3$ B. $-4i - 3$ C. $3i + 4$ D. $3i - 4$

32. Given $z = 2 + i$ and $w = 3 - i$, evaluate $w^2 z - 2$

A. $4(5 - i)$ B. $4(i - 5)$ C. $22 - 4i$ D. $22 + 4i$

33. Evaluate $\frac{1}{1+i}$

A. $\frac{1}{2}(1 + i)$ B. $\frac{1}{2}(1 - i)$ C. $\frac{1}{2}(i - 1)$ D. $-\frac{1}{2}(1 + i)$

34. Determine the modulus, argument of $i(2 - i)$.

A. $1.732,\ 63.43°$ B. $2.236,\ 63.43°$ C. $1.732,\ 26.57°$ D. $2.236,\ 26.57°$

35. The complex conjugate of $2 - i$ is

A. $2i - 1$ B. $-2i - 1$ C. $-i - 2$ D. $i + 2$

36. Given $Z_1 = 1 + 2i,\ Z_2 = 4 - 3i,\ Z_3 = -2 + 3i$, evaluate $|Z_1 + Z_2| - |Z_3| \times \sqrt{2}$

A. $\sqrt{10}$ B. 10 C. $2\sqrt{5}$ D. 0

37. Given $Z_1 = 1 + 2i,\ Z_2 = 4 - 3i,\ Z_3 = -2 + 3i$, evaluate $Z_1 + Z_2 - Z_3$

A. $7 - 4i$ B. $-5 + 8i$ C. $3 + 2i$ D. $1 - 2i$

38. Given $Z_1 = 1 + 2i,\ Z_2 = 4 - 3i,\ Z_3 = -2 + 3i$, evaluate $Z_2 - Z_1 + Z_3$

A. $7 - 4i$ B. $-5 + 8i$ C. $3 + 2i$ D. $1 - 2i$

39. If $x + iy = \frac{3-i}{2+i}$, find x and y

A. $x = 1, y = 1$ B. $x = -1, y = 1$ C. $x = 1, y = -1$ D. $x = -1, y = -1$

40. Evaluate $\frac{(2+i)^2(1-i)}{3i-1}$

A. $-\frac{2 + 11i}{5}$ B. $-\frac{4 + 22i}{5}$ C. $\frac{2 + 11i}{5}$ D. $\frac{4 + 22i}{5}$

41. If $Z = cos\theta + isin\theta$, then find the value of $\left(z + \frac{1}{z}\right)^2$ is

A. $2cos\theta$ B. $2isin\theta$ C. $4cos^2\theta$ D. $-4sin^2\theta$

42. Find the real part of $\frac{2+3i}{3+2i}$

A. $\frac{12}{13}$ B. $-\frac{5}{13}$ C. $\frac{5}{13}$ D. $-\frac{12}{13}$

43. Evaluate $(i + 3) - (2 - 3i) + (1 + 2i) - (2i + 1)$

A. $1 + 4i$ B. $1 - 4i$ C. $4 - i$ D. $4 + i$

44. Simplify $\left(\frac{3i-2}{1+2i}\right)$

A. $\frac{4+7i}{5}$ B. $\frac{4-7i}{25}$ C. $\frac{4-7i}{5}$ D. $\frac{4+7i}{25}$

45. Simplify $\left(\frac{1}{2-3i}\right)\left(\frac{1}{1+i}\right)$

A. $\frac{5i+1}{26}$ B. $\frac{5+i}{26}$ C. $\frac{5-i}{26}$ D. $\frac{5i-1}{26}$

46. Find the modulus of $1 - i\sqrt{3}$

A. $\sqrt{10}$ B. 4 C. $\sqrt{5}$ D. 2

47. Find the modulus and argument of $i - 1$

A. $\sqrt{2}, -45^o$ B. $\sqrt{2}, 45^o$ C. $\sqrt{2}, 135^o$ D. $2, -45^o$

48. Simplify $\frac{-1+3i}{2-i}$

A. $5 - 5i$ B. $5i - 5$ C. $1 - i$ D. $i - 1$

49. Polar complex number $4\angle\pi$ is equal to

A. $-4i$ B. $4i$ C. -4 D. 4

50. Solve the quadratic equation $x^2 - 4x = 8$

A. $4i$ B. $2i$ C. $2 + 2i$ D. $2i - 2$

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INEQUALITY

1. Find the solution set of the inequality: $3x + 2 \leq x + 4$

   A. $\{x : x \geq 1\}$ B. $\{x : x \leq 1\}$ C. $\{x : x \leq -1\}$ D. $\{x : x \geq -1\}$

2. Find the solution set of the inequality: $\frac{1}{2}x + 3 > 1 - x$

   A. $\left\{x : x < -\frac{3}{4}\right\}$ B. $\left\{x : x > -\frac{3}{4}\right\}$ C. $\left\{x : x < -\frac{4}{3}\right\}$ D. $\left\{x : x > -\frac{4}{3}\right\}$

3. Find the range of values of x for which the inequality $4x + 3 \geq 2x - 7$ is true.

   A. $x \geq -5$ B. $x \leq -5$ C. $x \geq 5$ D. $x \leq 5$

4. Determine the range of values of x for which the inequality $\frac{1}{4}x - 3 \leq 2 + x$ is true.

   A. $x \geq -\frac{3}{20}$ B. $x \geq -\frac{20}{3}$ C. $x \leq -\frac{3}{20}$ D. $x \leq -\frac{20}{3}$

5. Solve $2x + 1 \leq x + 5$

   A. $x \leq -4$ B. $x \leq 4$ C. $x \leq 2$ D. $x \leq -2$

6. Find the range of x for which $\frac{1}{3}x + 2 \geq \frac{1}{2}x + 1$

   A. $x \geq 3$ B. $x \geq -6$ C. $x \leq 6$ D. $x \leq -6$

7. Solve the inequality $3x - 2 \geq \frac{1}{2}x + 3$

   A. $x \geq -2$ B. $x \geq 2$ C. $x \leq -2$ D. $x \leq 2$

8. Find the range of x for which $4x + 1 \leq 3x + 2$

   A. $x \geq -1$ B. $x \geq 1$ C. $x \leq -1$ D. $x \leq 1$

9. Find the range of x for which $2 - 3x \geq x - 3$

   A. $x \leq -\frac{5}{4}$ B. $x \geq -\frac{5}{4}$ C. $x \leq \frac{5}{4}$ D. $x \geq \frac{5}{4}$

10. Solve $x + \frac{1}{2} \geq 2x + 5$

    A. $x \leq -\frac{9}{2}$ B. $x \geq -\frac{9}{2}$ C. $x \leq \frac{9}{2}$ D. $x \geq \frac{9}{2}$

11. Find the range of values of x for which the inequality $\frac{1}{3}x - 2 \leq \frac{2}{5}x + \frac{1}{4}$

    A. $x \geq -\frac{4}{135}$ B. $x \geq -\frac{135}{4}$ C. $x \geq \frac{4}{135}$ D. $x \geq \frac{135}{4}$

12. Find the range of values of x for which the inequality $4x + \frac{1}{3} \geq \frac{2}{3}x - 2$

    A. $x \leq \frac{7}{10}$ B. $x \leq -\frac{7}{10}$ C. $x \geq \frac{7}{10}$ D. $x \geq -\frac{7}{10}$

13. Find the set solution to the inequality $3x - 4 < 2x + 1 \leq 3x + 2$

    A. $1 \leq x < 5$ B. $-1 < x \leq 5$ C. $-1 \leq x \leq 5$ D. $-1 \leq x < 5$

BRAINSTORM GROUP

14. Find the set solution to the inequality $2x + 4 < x + 3 < 4x - 5$

A. $\left\{x : x < -1 \cup x > 2\frac{2}{3}\right\}$
B. $\left\{x : -1 < x < 2\frac{2}{3}\right\}$
C. $\left\{x : x > -1 \cup x < 2\frac{2}{3}\right\}$
D. $\left\{x : -1 > x > 2\frac{2}{3}\right\}$

15. Find the solution set of the inequality: $3x - 2 < x + 1 \leq 2x + 3$

A. $2 \leq x < -\frac{3}{2}$
B. $2 < x \leq -\frac{3}{2}$
C. $-2 < x \leq \frac{3}{2}$
D. $-2 \leq x < \frac{3}{2}$

16. Solve the inequality $x^2 + x - 6 > 0$

A. $\{x: -3 < x < 2\}$
B. $\{x: -2 < x < 3\}$
C. $\{x: x < -3\} \cup \{x: x > 2\}$
D. $\{x: x < -3\} \cup \{x: x > 2\}$

17. Solve the inequality $x^2 + 3x - 4 \leq 0$

A. $\{x: -3 \leq x \leq 1\}$
B. $\{x: -4 \leq x \leq 1\}$
C. $\{x: 4 \leq x \leq 1\}$
D. $\{x: -4 \leq x \leq -1\}$

18. Find the range of values of x for which $2x^2 + 7x - 15 > 0$

A. $\left\{x: x < 5 \text{ or } x > \frac{3}{2}\right\}$
B. $\left\{x: x < 5 \text{ or } x > -\frac{3}{2}\right\}$
C. $\left\{x: x < -5 \text{ or } x > \frac{3}{2}\right\}$
D. $\left\{x: x < -5 \text{ or } x > \frac{3}{2}\right\}$

19. Solve $x^2 - 2x - 8 > 0$

A. $\{x: x < 2 \text{ or } x > 4\}$
B. $\{x: x < -2 \text{ or } x > -4\}$
C. $\{x: x < -2 \text{ or } x > 4\}$
D. $\{x: x < 2 \text{ or } x > -4\}$

20. Solve $x^2 - 4 \geq 0$

A. $\{x: x < 0 \text{ or } x > 2\}$
B. $\{x: x < -2 \text{ or } x > 2\}$
C. $\{x: 0 < x < 2\}$
D. $\{x: -2 < x < 2\}$

21. Solve the inequality $3x^2 + 10x - 8 < 0$

A. $\left\{x: -4 < x < \frac{3}{2}\right\}$
B. $\left\{x: -2 < x < \frac{2}{3}\right\}$
C. $\left\{x: -2 < x < \frac{4}{3}\right\}$
D. $\left\{x: -4 < x < \frac{2}{3}\right\}$

22. Find the range of values for which $\frac{1}{x-3} > 1$

A. $\{x: -3 < x < -4\}$
B. $\{x: 3 < x < 4\}$
C. $\{x: 3 < x < -4\}$
D. $\{x: -3 < x < 4\}$

23. Solve the inequality $x^2 - x - 10 < 2$

A. $\{x: -3 < x < 4\}$
B. $\{x: -3 < x < -4\}$
C. $\{x: -3 < x < -4\}$
D. $\{x: 3 < x < 4\}$

24. Solve the inequality $3 - 5x - 2x^2 > 0$

A. $\left\{x: -2 < x < \frac{1}{3}\right\}$ B. $\left\{x: -3 < x < -\frac{1}{2}\right\}$

C. $\left\{x: -3 < x < \frac{1}{2}\right\}$ D. $\left\{x: 3 < x < \frac{1}{2}\right\}$

25. Solve the inequality $8 + 10x - 3x^2 \leq 0$

A. $\left\{x: x \leq -\frac{2}{3}\right\} \cup \{x; x \geq 4\}$ B. $\left\{x: x \leq -\frac{2}{3}\right\} \cup \{x; x \geq -4\}$

C. $\left\{x: x \leq \frac{2}{3}\right\} \cup \{x; x > 4\}$ D. $\left\{x: x \leq -\frac{2}{3}\right\} \cup \{x; x > 4\}$

26. Solve the inequality $x^2 + 4x - 5 > 0$

A. $\{x: x < 5\} \cup \{x; x > -1\}$ B. $\{x: x < -5\} \cup \{x; x > 1\}$

C. $\{x: x < 5\} \cup \{x; x > 1\}$ D. $\{x: x < -5\} \cup \{x; x > -1\}$

27. Solve the inequality $x^2 - 4x - 8 < 2 - x$

A. $\{x: -2 < x < -5\}$ B. $\{x: 2 < x < 5\}$

C. $\{x: 2 < x < -5\}$ D. $\{x: -2 < x < 5\}$

28. Solve $|2x - 3| < 4$

A. $-\frac{1}{2} < x < -\frac{7}{2}$ B. $-\frac{1}{2} > x < \frac{7}{2}$

C. $-\frac{1}{2} > x > \frac{7}{2}$ D. $-\frac{1}{2} < x < \frac{7}{2}$

29. Solve the inequality $|3x - 2| > 4$

A. $\left\{x: x < -\frac{2}{3}\right\} \cup \{x: x > 2\}$ B. $\left\{x: x < \frac{2}{3}\right\} \cup \{x: x > 2\}$

C. $\left\{x: x > -\frac{2}{3}\right\} \cup \{x: x > 2\}$ D. $\left\{x: x > -\frac{2}{3}\right\} \cup \{x: x > 2\}$

30. Solve the inequality $|x - 2| < |x + 3|$

A. $\left\{x: x < -\frac{1}{2}\right\}$ B. $\left\{x: x > \frac{1}{2}\right\}$ C. $\left\{x: x > -\frac{1}{2}\right\}$ D. $\left\{x: x < \frac{1}{2}\right\}$

31. Solve the inequality $\frac{3}{4}(1 - x) > \frac{1}{4} - x$

A. $x > 2$ B. $x > -2$ C. $x < 2$ D. $x < -2$

32. Solve $|5x + 5| - 8 \leq 17$

A. $-5 \leq x \leq 5$ B. $-5 \leq x \leq 4$ C. $0 \leq x \leq 4$ D. $-6 \leq x \leq 4$

33. Which of the following represents the solution set of the inequality $2x + 1 \leq x + 5$

A. $x \leq -4$ B. $x \leq 4$ C. $x \geq -4$ D. $x \geq 4$

34. Which of the following represents the solution set of the inequality $\frac{1}{3}x + 2 \geq \frac{1}{2}x + 1$

A. $x \leq -6$ B. $x \leq 6$ C. $x \geq -6$ D. $x \geq 6$

35. Find the solution set of the inequality $x^2 + 3x - 4 \leq 0$

A. $4 \leq x \leq -1$ B. $-4 \leq x \leq -1$ C. $-4 \leq x \leq 1$ D. $4 \leq x \leq 1$

36. Determine the solution set of the inequality $x^2 - x - 10 < 2$

A. $\{x : 3 < x < 4\}$ B. $\{x : -4 < x < 3\}$ C. $\{x : -3 < x < 4\}$ D. $\{x : -4 < x < -3\}$

37. Find the solution set of the inequality $x^2 - 2x - 3 \geq 0$

A. $x \leq 1$ *or* $x \geq 3$ B. $x \leq -3 or\ x \geq 1$ C. $x \leq -1$ *or* $x \geq 3$ D. $x \leq -3$ *or* $x \geq 1$

38. Determine the solution set of the inequality $3 - 5x - 2x^2 > 0$

A. A. $\{x : \frac{1}{2} < x < 3\}$ B. $\{x : -3 < x < -\frac{1}{2}\}$ C. $\{x : -3 < x < \frac{1}{2}\}$ D. $\{x : -\frac{1}{2} < x < 3\}$

39. Find the solution set of the inequality $8 + 10x - 3x^2 \leq 0$

A. $\left\{x: x < -\frac{2}{3}\right\} \cup \{x: x > 4\}$ B. $\left\{x: x < \frac{2}{3}\right\} \cup \{x: x > 4\}$

C. $\{x: x < -4\} \cup \left\{x: x > \frac{2}{3}\right\}$ D. $\{x: x < -4\} \cup \left\{x: x > \frac{2}{3}\right\}$

40. Find the solution set of the inequality $x^2 + 4x - 5 > 0$

A. $\{x: x < -5\} \cup \{x: x > -1\}$ B. $\{x: x < -5\} \cup \{x: x > 1\}$

C. $\{x: x < -1\} \cup \{x: x > 5\}$ D. $\{x: x < 1\} \cup \{x: x > 5\}$

41. Find the solution set of the inequality $x^2 - 4x - 8 < 2 - x$

A. $\{x : -2 < x < 5\}$ B. $\{x : 2 < x < 5\}$ C. $\{x : -5 < x < 2\}$ D. $\{x : -5 < x < -2\}$

42. Determine the values of x satisfying: $|x| < 5$

A. $-5 < x < 5$ B. $x < -5, x > 5$ C. $x > -5$ D. $x < 5$

43. Determine the values of x satisfying: $|x + 3| \geq 7$

A. $x \leq -4, x \geq -10$ B. $x \leq -10, x \geq -4$ C. $x \leq -4, x \geq 10$ D. $x \leq -10, x \geq 4$

44. Find the solution set of the inequality $3x - 2 < x + 1 \leq 2x + 3$

A. $\frac{3}{2} < x \leq 2$ B. $\frac{3}{2} \leq x < 2$ C. $-2 < x \leq \frac{3}{2}$ D. $-2 \leq x < \frac{3}{2}$

45. Determine the values of x satisfying: $|x + 3| \geq 4$

A. $x \leq -7, x \geq 1$ B. $x \leq -7, x \geq 1$ C. $x \leq -7, x \geq 1$ D. $x \leq -7, x \geq 1$

46. Determine the values of x satisfying: $|x - 4| \leq 1$

A. $-3 \leq x \leq 5$ B. $3 \leq x \leq 5$ C. $-5 \leq x \leq 3$ D. $-5 \leq x \leq -3$

47. Find the solution set of the inequality $2x - 3 < x + 7$

A. $x > -10$ B. $x < -10$ C. $x < 10$ D. $x > 10$

48. Find the solution set of the inequality $3x + 4 > 1 - 2x$

A. $x > \frac{3}{5}$ B. $x < \frac{3}{5}$ C. $x > -\frac{3}{5}$ D. $x < -\frac{3}{5}$

49. Find the solution set of the inequality $\frac{1}{3}x - 2 \leq \frac{2}{5}x + \frac{1}{4}$A. $x \geq -\frac{135}{4}$

A. $x \geq \frac{135}{4}$ B. A. $x \geq -\frac{135}{4}$ C. A. $x \leq \frac{135}{4}$ D. A. $x \leq -\frac{135}{4}$

50. Find the solution set of the inequality $4x + \frac{1}{3} \geq \frac{2}{3}x - 2$ A. $x \geq -\frac{7}{10}$

A. $x \geq -\frac{7}{10}$ B. $x \leq -\frac{7}{10}$ C. $x \geq \frac{7}{10}$ D. $x \leq \frac{7}{10}$

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SURD

1. Solve $\frac{\sqrt{4\times25\times3}}{\sqrt{9\times4}}$

   A. $\frac{4\sqrt{3}}{3}$ B. $\frac{5\sqrt{3}}{3}$ C. $5\sqrt{3}$ D. $4\sqrt{3}$

2. Simplify $\sqrt{20}+\sqrt{45}+\sqrt{125}-2\sqrt{80}$

   A. $2\sqrt{5}$ B. $3\sqrt{5}$ C. $4\sqrt{5}$ D. $5\sqrt{5}$

3. Simplify $3\sqrt{2}-\sqrt{32}+\sqrt{50}+\sqrt{98}$

   A. $5\sqrt{2}$ B. $6\sqrt{2}$ C. $10\sqrt{2}$ D. $11\sqrt{2}$

4. The conjugate of $\sqrt{3}-\sqrt{2}$ is

   A. $\sqrt{2}-\sqrt{3}$ B. $\sqrt{2}+\sqrt{3}$ C. $\sqrt{3}-\sqrt{2}$ D. $\sqrt{3}+\sqrt{2}$

5. The conjugate of $2\sqrt{5}+3\sqrt{3}$ is

   A. $2\sqrt{5}-3\sqrt{3}$ B. $2\sqrt{5}+3\sqrt{3}$ C. $2\sqrt{5}-\sqrt{3}$ D. $2\sqrt{5}+\sqrt{3}$

6. Simplify $(3\sqrt{2}-1)(3\sqrt{2}+1)$

   A. 18 B. 17 C. 13 D. 12

7. Simplify $(7\sqrt{2}+\sqrt{3})(7\sqrt{2}-\sqrt{3})$

   A. 80 B. 85 C. 95 D. 101

8. Express $\frac{8-\sqrt{18}}{\sqrt{2}}$ in terms of $a+b\sqrt{2}$

   A. $-3+4\sqrt{2}$ B. $3+4\sqrt{2}$ C. $-3-4\sqrt{2}$ D. $3-4\sqrt{2}$

9. Simplify $5\sqrt{12}-4\sqrt{75}+3\sqrt{48}$

   A. $\sqrt{3}$ B. $2\sqrt{3}$ C. $3\sqrt{2}$ D. $4\sqrt{3}$

10. Simplify $3\sqrt{18}-\frac{16}{\sqrt{2}}+\sqrt{50}$

    A. $6\sqrt{2}$ B. $2\sqrt{6}$ C. $4\sqrt{2}$ D. $3\sqrt{2}$

11. The square of $3-\sqrt{2}$ is

    A. $6\sqrt{2}$ B. 7 C. $13-6\sqrt{2}$ D. $11-6\sqrt{2}$

12. Simplify $(\sqrt{5}+2)(\sqrt{5}-2)$

    A. 1 B. 3 C. $2\sqrt{5}$ D. $9+4\sqrt{5}$

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13. Simplify $\frac{1}{2-\sqrt{3}} - \frac{1}{2+\sqrt{3}}$

A. 4 B. $4\sqrt{3}$ C. $2\sqrt{3}$ C. 2

14. Simplify $\frac{\sqrt{8}-\sqrt{2}}{\sqrt{8}+\sqrt{2}}$

A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{2}{3}$ D. $\sqrt{2}$

15. Simplify $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$

A. $\frac{1}{2}\sqrt{15}$ B. $8 + 2\sqrt{15}$ C. $4 + \sqrt{15}$ D. $4 - \sqrt{15}$

16. Simplify $2 - \frac{1}{\sqrt{2}-1}$

A. $\sqrt{2}$ B. $1 - \sqrt{2}$ C. $\sqrt{2} - 1$ D. $2\sqrt{2} + 1$

17. Simplify $\frac{3\sqrt{15} \times 2\sqrt{48}}{\sqrt{32} \times 4\sqrt{45}}$

A. $\frac{3\sqrt{2}}{4}$ B. $\frac{3}{4}$ C. $\frac{2\sqrt{3}}{3}$ D. $\frac{2\sqrt{2}}{3}$

18. Simplify $(\sqrt{3} - \sqrt{2})^2(\sqrt{3} + \sqrt{2})^2$

A. 0 B. 1 C. $\sqrt{6}$ D. 6

19. Find the value of a and b if $a\sqrt{5} + b\sqrt{2}$ is a square root $95 - 30\sqrt{10}$

A. $a = 5, b = 2$ B. $a = 2, b = -5$ C. $a = -5, b = 3$ D. $a = 3, b = -5$

20. Simplify $\sqrt{3} + 2 - \frac{1}{2-\sqrt{3}}$

A. $\sqrt{3} - 1$ B. $2\sqrt{3}$ C. $\sqrt{3}$ D. 0

21. Solve $5\sqrt{27} - 3\sqrt{12} - 4\sqrt{3}$

A. $2\sqrt{3}$ B. $3\sqrt{3}$ C. $4\sqrt{3}$ D. $5\sqrt{3}$

22. Solve $2\sqrt{135} - 2\sqrt{60} + \sqrt{15} - \sqrt{240}$

A. $-\sqrt{5}$ B. $-\sqrt{15}$ C. $\sqrt{5}$ D. $\sqrt{15}$

23. Solve $6 + \sqrt{27} + \sqrt{75}$

A. $6 + 10\sqrt{3}$ B. $6 + 8\sqrt{3}$ C. $6 + 6\sqrt{3}$ D. $6 + 4\sqrt{3}$

24. Solve $\sqrt{52} - \sqrt{117} + 4\sqrt{13}$

A. $3\sqrt{13}$ B. $5\sqrt{13}$ C. $2\sqrt{13}$ D. $\sqrt{13}$

25. Simplify $\sqrt{224} - \sqrt{126} - \sqrt{56}$

A. $\sqrt{14}$ B. $-\sqrt{7}$ C. $\sqrt{7}$ D. $-\sqrt{14}$

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26. Express $\frac{\sqrt{18}\times\sqrt{20}\times\sqrt{24}}{\sqrt{8}\times\sqrt{30}}$ in basic form

A. $\sqrt{6}$ B. $-\sqrt{6}$ C. -6 D. 6

27. Express $\frac{\sqrt{3}\times\sqrt{18}\times\sqrt{39}}{\sqrt{24}\times\sqrt{26}}$ in basic form

A. $\frac{3\sqrt{6}}{2}$ B. $\frac{3\sqrt{6}}{4}$ C. $\frac{3\sqrt{3}}{2}$ D. $\frac{3\sqrt{3}}{4}$

28. Simplify $\sqrt{3}+\frac{1}{\sqrt{3}}-\frac{1}{\sqrt{27}}$

A. $\frac{9\sqrt{3}}{3}$ B. $\frac{13\sqrt{3}}{6}$ C. $\frac{11\sqrt{3}}{9}$ D. $\frac{11\sqrt{3}}{3}$

29. Simplify $2\sqrt{2}-\frac{3}{\sqrt{2}}+\frac{4}{\sqrt{8}}$

A. $3\sqrt{2}$ B. $4\sqrt{2}$ C. $\frac{\sqrt{2}}{2}$ D. $\frac{3\sqrt{2}}{2}$

30. Rationalize $\frac{12}{\sqrt{24}-\sqrt{6}}$

A. $2\sqrt{6}$ B. $2\sqrt{3}$ C. $4\sqrt{6}$ D. $4\sqrt{3}$

31. Simplify $(\sqrt{6}-\sqrt{5})(\sqrt{8}+\sqrt{15})$

A. $\sqrt{10}-\sqrt{3}$ B. $\sqrt{10}+\sqrt{3}$ C. $-\sqrt{10}-\sqrt{3}$ D. $-\sqrt{10}+\sqrt{3}$

32. Simplify $(\sqrt{5}+\sqrt{15})(2\sqrt{3}-1)$

A. $\sqrt{5}+\sqrt{15}$ B. $\sqrt{5}-\sqrt{15}$ C. $5\sqrt{5}+\sqrt{15}$ D. $5\sqrt{5}-\sqrt{15}$

33. Simplify $\left(\sqrt{2}+\frac{1}{\sqrt{2}}\right)^2$

A. $2\frac{1}{2}$ B. $4\frac{1}{2}$ C. 2 D. 4

34. Simplify $(3\sqrt{2}-2\sqrt{3})(5\sqrt{2}-\sqrt{3})$

A. $36-11\sqrt{6}$ B. $24-13\sqrt{6}$ C. $36-13\sqrt{6}$ D. $24+13\sqrt{6}$

35. Simplify $(3\sqrt{2}-\sqrt{5})^2$

A. $13+\sqrt{10}$ B. $13+6\sqrt{10}$ C. $23-\sqrt{10}$ D. $23-6\sqrt{10}$

36. Rationalize $\frac{1}{2-\sqrt{3}}$

A. $2-\sqrt{3}$ B. $2+\sqrt{3}$ C. $-2+\sqrt{3}$ D. $-2-\sqrt{3}$

37. Rationalize $\frac{4}{3+\sqrt{7}}$

A. $2(3+\sqrt{7})$ B. $3(3-\sqrt{7})$ C. $2(3-2\sqrt{7})$ D. $2(3-\sqrt{7})$

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38. Rationalize $\frac{\sqrt{6}+\sqrt{2}}{\sqrt{6}-\sqrt{2}}$

A. $2-\sqrt{3}$ B. $2+\sqrt{3}$ C. $-2+\sqrt{3}$ D. $-2-\sqrt{3}$

39. Rationalize $\frac{\sqrt{7}+3\sqrt{3}}{2\sqrt{7}+\sqrt{3}}$

A. $\frac{\sqrt{21}}{5}$ B. $\frac{1+\sqrt{21}}{5}$ C. $\frac{7+\sqrt{21}}{5}$ D. $\frac{1-\sqrt{21}}{5}$

40. Simplify $\sqrt{90} \times \sqrt{3.6}$

A. $\sqrt{18}$ B. 6 C. $\sqrt{6}$ D. 18

41. $\sqrt{64} - \sqrt[3]{64} = \sqrt[4]{y}$, where $y > 0$, the value of y is

A. 256 B. 16 C. 625 D. 81

42. What is the square root of $(8 + 2\sqrt{15})$?

A. $\sqrt{5} + \sqrt{3}$ B. $2\sqrt{5} + 2\sqrt{3}$ C. $2\sqrt{2} + 2\sqrt{6}$ D. $\sqrt{2} + \sqrt{6}$

43. What is the square root of $(10 + \sqrt{25})(12 - \sqrt{49})$?

A. $2\sqrt{3}$ B. $3\sqrt{3}$ C. $4\sqrt{3}$ D. $5\sqrt{3}$

44. Find the value of k if $(3 + 2\sqrt{5})^2 = 29 + k\sqrt{5}$

A. 6 B. 29 C. 12 D. 39

45. If $\sqrt{x} - \sqrt{x-2} = 1$, find the positive values of x which satisfies the equation.

A. $\frac{3}{4}$ B. $\frac{5}{4}$ C. $\frac{7}{4}$ D. $\frac{9}{4}$

46. Simplify $\sqrt{27} + \sqrt{12} - \sqrt{48}$

A. $2\sqrt{3}$ B. $-\sqrt{3}$ C. $-3\sqrt{3}$ D. $\sqrt{3}$

47. Solve the equation $\sqrt{x+8} + \sqrt{x+1} = 7$

A. 2 B. 8 C. 7 D. 20

48. Solve the equation $2\sqrt{3x+4} + x = 36$

A. 64 B. 32 C. 16 D. 20

49. Given $\frac{8-3\sqrt{6}}{2\sqrt{3}+3\sqrt{2}}$ is equal to $m\sqrt{3} + n\sqrt{2}$, where m and n are rational numbers. Find n.

A. $-\frac{17}{2}$ B. -17 C. 7 D. $\frac{7}{2}$

50. Express $\frac{1}{\sqrt{5}+\sqrt{3}}$ in the form $p\sqrt{5} + q\sqrt{3}$, where p and q are rational numbers.

A. $\frac{1}{2}\sqrt{5} + \frac{1}{2}\sqrt{3}$ B. $\frac{1}{2}\sqrt{5} - \frac{1}{2}\sqrt{3}$ C. $-\frac{1}{2}\sqrt{5} + \frac{1}{2}\sqrt{3}$ D. $-\frac{1}{2}\sqrt{5} - \frac{1}{2}\sqrt{3}$

INDICES AND LOGARITHMS

1. $2^{y-x} = 16 \; and \; 4^{y+x} = 8$, find the value of x + y

   A. 3 B. $\frac{3}{2}$ C. -3 D. $-\frac{3}{2}$

2. $Find \; x \; in \; log_{10}(x-9) + \; log_{10}x = 1$

   A. 1 B. -1 C. 2 D. -10

3. $log_{10}(x-14) + \; log_{10}(x-5) = 1$

   A. $4 \; or \; 15$ B. $4 \; or \; -15$ C. $-4 \; or \; 15$ D. $-4 \; or \; -15$

4. $3 \times 5^x = {}^{3}/_{125}, what \; is \; the \; value \; of \; x?$

   A. $-\frac{1}{3}$ B. $\frac{1}{3}$ C. 3 D. -3

5. ${}^{32^{2}/_{5}} \times {}^{64^{1}/_{18}}$

   A. $\sqrt[3]{256}$ B. $\sqrt[3]{128}$ C. $\sqrt[3]{64}$ D. $\sqrt[3]{16}$

6. $Find \; M \; given \; logM = \frac{1}{4}log81$

   A. $-\frac{1}{3}$ B. $\frac{1}{3}$ C. 3 D. -3

7. $log_x(8x-3) - \; log_x 4 = 2$

   A. $-\frac{1}{2} \; or \; -\frac{3}{2}$ B. $-\frac{1}{2} \; or \; \frac{3}{2}$ C. $-\frac{1}{2} \; or \; \frac{3}{2}$ D. $\frac{1}{2} \; or \; \frac{3}{2}$

8. $log_3 27 - \; log_3 9 + \; log_3 81$

   A. 5^3 B. 3^5 C. 5 D. -5

9. $log_a 36 + \frac{1}{2}log_a 256 - \; 2log_a 48$

   A. $-log_a 4$ B. $log_a 16$ C. $log_a 8$ D. $-4log_a 2$

10. $Find \; the \; value \; for \; x \; in \; 4^x - 7 \times 2^x + 12 = 0$

    A. 2 B. 3 C. 4 D. 5

11. Find the value of r that satisfies the equation $2r^{-3} = -16$

    A. $\frac{1}{2}$ B. $-\frac{1}{2}$ C. $\frac{1}{4}$ D. $-\frac{1}{4}$

12. Given $5x = 40x^{-1/2}$, find x

    A. 2 B. -2 C. 4 D. 8

13. Simplify $\frac{log0.2}{log25}$

    A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. -2 D. 2

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14. Evaluate $\frac{log8-log4}{log4-log2}$

A. -1 B. 1 C. -2 D. 2

15. Solve the equation $2^{x^2} = \frac{1}{4}8^x$

A. $\frac{1}{2}$ *and* 2 B. $\frac{1}{2}$ *and* 1 C. 1 *and* 2 D. -1 *and* -2

16. Evaluate $log_3 27 \times log_{1/2} 32$

A. -15 B. 15 C. -30 D. 30

17. If $logx - \log(2x-1) = 1$, find x

A. $\frac{19}{10}$ B. $-\frac{19}{10}$ C. $-\frac{10}{19}$ D. $\frac{10}{19}$

18. Evaluate $log_2\sqrt{8} + log_{1/3}\sqrt{3} - log_4 8$

A. $-\frac{1}{2}$ B. $\frac{3}{2}$ C. $\frac{1}{2}$ D. $-\frac{3}{2}$

19. Given that $log_3(x-y) = 1$ and $log_3(2x+y) = 2$, find the value of $2y - x$

A. -1 B. -2 C. 2 D. 4

20. Solve the simultaneous equations $log_2 x - log_2 y = 2,\ log_2(x-2y) = 3$

A. $x = 18, y = \frac{9}{2}$ B. $x = 9, y = \frac{9}{2}$ C. $x = \frac{9}{2}, y = 18$ D. $x = 9, y = -\frac{9}{2}$

21. Solve the equations $2log_{\sqrt{2}} 16 = x$

A. 32 B. 16 C. 8 D. 4

22. Simplify: $\frac{9^{-1/2} \times 81^{3/4}}{9^{1/2}}$

A. $\frac{1}{9}$ B. $\frac{1}{3}$ C. 3 D. 9

23. Find the value of x if $4^{(x-1/2)} = 2^{x^2}$

A. $\frac{1}{2}$ B. 1 C. 2 D. 3

24. Simplify: $(0.064)^{-1/3}$

A. 2 B. $2\frac{1}{2}$ C. 4 D. $\frac{1}{4}$

25. Simplify: $\sqrt{\frac{54y^{-4}}{6y^{-8}}}$

A. $3y^{-2}$ B. $3y^4$ C. $3y^2$ D. $9y^4$

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26. Solve for x: $5(2^{x-3}) = 160$

A. 6 B. 7 C. 8 D. 10

27. Simplify: $\sqrt[3]{(729v^{-6})^{1/2}}$

A. $\frac{1}{3v}$ B. $3v$ C. $\frac{v}{3}$ D. $\frac{3}{v}$

28. Solve for z: $9^{2z+1} = \frac{81^{z+2}}{3^z}$

A. 2 B. 4 C. 6 D. -4

29. If $3^{2x+1} = 243$, find x

A. -2 B. 2 C. -4 D. 4

30. Simplify: $\frac{3^{2c+1} \times 9^c \times 4^{2c}}{18^c \times 2^c \times 6^{2c}}$

A. 3^c B. 2^c C. 1 D. 3

31. Evaluate $log_3 27 + log_3 6 - log_3 54$

A. B. $log_3 2$ C. $\frac{1}{3}$ D. 1

32. If $2log_4 p + log_4 q^2 = 0$, then p is given by

A. $^2/_q$ B. q C. $^q/_2$ D. $^1/_q$

33. The value $4^{3/2} + log_5 0.04 + log_3 81$ is

A. 2 B. 8 C. 10 D. 12

34. Solve for x if $5\log(x+3) = log 32$

A. -1 B. 0 C. 1 D. 2

35. Find c if $log_2 c = log_9 3$

A. $^2/_3$ B. $^1/_2$ C. $\sqrt{2}$ D. 4

36. Evaluate $\frac{log_2 8 - log_{1/3} 81}{log_4 \frac{1}{64}}$

A. $-\frac{7}{3}$ B. $\frac{7}{3}$ C. $-\frac{3}{2}$ D. $\frac{2}{3}$

37. Simplify $log_5 25^x - log_5(0.04)$

A. $2x + 1$ B. $2x + 2$ C. $^x/_2$ D. $2x - 2$

38. Find the value of ab if $a = log_2 3$ and $b = log_3 8$

A. 2 B. 3 C. 4 D. 8

39. Evaluate $\frac{log_{10}12 - log_{10}3}{log_{10}8}$

A. $4\frac{1}{2}$ B. $\frac{9}{8}$ C. $\frac{2}{3}$ D. $\frac{1}{2}$

40. The value of $(4^{1/4})^6$ is

A. $^1/_8$ B. 4 C. 6 D. 8

41. Evaluate $256^{0.16} \times 256^{0.09}$

A. 4 B. 16 C. 64 D. 256

42. Expresses the function $y = log_3 9$ in exponential form?

A. $y = 2$ B. $y = 9^3$ C. $3^y = 9$ D. $9^y = 3$

43. Evaluate $\frac{1}{2}log_4 8 + log_4 32 - log_4 2$

A. $\frac{11}{4}$ B. $\frac{5}{2}$ C. $\frac{9}{4}$ D. $\frac{9}{2}$

44. Solve for x: $logx = \frac{1}{3}log27$

A. $\frac{1}{3}$ B. $\frac{1}{9}$ C. 3 D. 9

45. Express as a single logarithm $\frac{1}{2}[(2lna + lnb) - 5lnc]$

A. $\frac{2ab}{5c}$ B. $\frac{a^2b}{c^5}$ C. $ln\sqrt{\frac{2ab}{5c}}$ D. $ln\sqrt{\frac{a^2b}{c^5}}$

46. Given: $f(x) = 3log_6(2x - 1)$, find $f(7)$

A. 4.293 B. 4.294 C. 4.295 D. 4.296

47. Solve the following equation $log_2(x - 3) + log_2 x = 2$

A. -1 *and* 4 B. -1 *and* -4 C. 1 *and* -4 D. 1 *and* 4

48. Simplify $log_9 3 + log_9 243 - 2log_9 3$

A. 2 B. 3 C.9 D. 81

49. Simplify $log_5 12.5 + log_5 5$

A. 1.696 B. 1.697 C. 1.698 D. 1.699

50. Simplify $log_2\sqrt{8} + log_3\sqrt{3}$

A. $\frac{1}{2}$ B. $\frac{3}{2}$ C. 2 D. 4

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MATHEMATICAL INDUCTION

1. Which of the following represents the sum of n natural numbers

   A. $\frac{1}{2}n(n+1)$ B. $\frac{1}{6}n(n+1)(2n+1)$ C. n^2 D. $\frac{1}{4}n^2(n+1)^2$

2. Which of the following represents the sum of squares of n natural numbers

   A. $\frac{1}{2}n(n+1)$ B. $\frac{1}{6}n(n+1)(2n+1)$ C. n^2 D. $\frac{1}{4}n^2(n+1)^2$

3. Find the value of $1^2 + 2^2 + \ldots + 10^2$ A.

   A. 375 B. 385 C. 395 D. 365

4. Find the value of $11^2 + 12^2 + \ldots + 20^2$

   A. 2870 B. 1435 C. 2485 D. 4970

5. Find the value of $1^2 + 2^2 + \ldots + 15^2$

   A. 385 B. 620 C. 2480 D. 1240

6. Find the value of $1^3 + 2^3 + \ldots + 10^3$

   A. 385 B. 2485 C. 2025 D. 3025

7. Evaluate $\sum_{r=1}^{5} r^2$

   A. 45 B. 55 C. 60 D. 110

8. If the square sum of n consecutive numbers is 385. Find n if the least of the numbers is 1.

   A. 5 B. 10 C. 15 D. 20

9. Evaluate $\sum_{r=1}^{4} r(r^2+1)$

   A. 100 B. 110 C. 120 D. 130

10. Evaluate $\frac{(n+2)!}{n!}$

    A. $(n+1)(n+2)$ B. $\frac{1}{2}(n+1)(n+2)$ C. $(n-1)(n-2)$ D. $n(n-1)(n-2)$

11. Which of the following represents the sum of cubes of n natural

    A. $\frac{1}{2}n(n+1)$ B. $\frac{1}{6}n(n+1)(2n+1)$ C. n^2 D. $\frac{1}{4}n^2(n+1)^2$

12. Which of the following represents the sum of odd n natural numbers

    A. $\frac{1}{2}n(n+1)$ B. $\frac{1}{6}n(n+1)(2n+1)$ C. n^2 D. $\frac{1}{4}n^2(n+1)^2$

13. Obtain the sum of positive even integers less than 100.

    A. 4950 B. 5050 C. 5500 D. 2450

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14. By induction hypothesis, the series $1^2 + 2^2 + \ldots + k^2$ can be proved equivalent to

A. $\frac{k^2+2}{4}$ B. $\frac{k(k+1)(2k+1)}{6}$ C. $\frac{k(k+1)}{4}$ D. $k^2 + k$

15. For every natural number n, $n(n+1)$ is always

A. *even* B. *odd* C. *multiple of* 3 D. *multiple of* 4

16. For principle of mathematical induction to be true, what type of number should n be?

A. Whole number B. Natural number
C. Rational number D. Any form of number

17. If the square sum of n consecutive numbers is 1240. Find n if the least of the numbers is 1.

A. 5 B. 10 C. 15 D. 20

18. Evaluate $\sum_{r=1}^{10} r^2$

A. 375 B. 385 C. 395 D. 365

19. Evaluate $\sum_{r=1}^{5} r^3 + 1$

A. 220 B. 225 C. 230 D. 235

20. Evaluate $\sum_{r=1}^{5} r(r+1)$

A. 56 B. 55 C. 70 D. 85

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BINOMIAL EXPANSION

1. Obtain the first three terms of the expansion $(1+x)^{-m}$.

   A. $1-mx-\frac{1}{2}m(m+1)x^2$ B. $1+mx+\frac{1}{2}m(m-1)x^2$

   C. $1-mx+\frac{1}{2}m(m+1)x^2$ D. $1+mx+\frac{1}{2}m(m-1)x^2$

2. Obtain the fourth term of the expansion $(2+3x)^{11}$

   A. $1140480x^2$ B. $1104480x^4$ C. $1140480x^3$ D. $1140480x^5$

3. Find the first three terms of the expansion $(1+x^2)^{-2}$

   A. $1-2x^2+3x^4$ B. $1-2x^2-3x^4$ C. $1+2x^2-3x^4$ D. $1+2x^2+3x^4$

4. If $(1-2x)^6 = 1+Px+Qx^2$ ... find the values of the constants P and Q.

   A. $60\ and\ -12$ B. $-60\ and\ 12$ C. $12\ and\ -60$ D. $-12\ and\ 60$

5. What is the coefficient of x^2 in $(1+x)^{-2}$.

   A. 2 B. -2 C. 3 D. -3

6. Find the term independent of x in the expansion of $(x^2-\frac{2}{x})^6$.

   A. -16 B. 16 C. -240 D. 240

7. What is the coefficient of x^3 in $(2+3x)^4$.

   A. 72 B. 108 C. 180 D. 216

8. Find the term independent of x in the expansion of $(x-\frac{3}{x})^8$.

   A. 5760 B. 5670 C. -5670 D. -5760

9. Calculate the coefficient of x^3y^2 in $(2x-3y)^5$.

   A. 10 B. -108 C. 720 D. -720

10. What is the coefficient of x^2 in $(1+x)^{1/3}$.

    A. $-\frac{1}{3}$ B. $\frac{1}{3}$ C $-\frac{1}{9}$ D. $\frac{1}{9}$

11. Obtain the seventh term of the expansion of $(2\ +\ 3x)^{11}$.

    A. $330\times 2^4\times 3^7\times x^7$ B. $462\times 2^5\times 3^6\times x^6$

    C. $165\times 2^4\times 3^7\times x^7$ D. $330\times 2^5\times 3^6\times x^6$

12. Obtain the fourth term of the expansion of $(2\ +\ 3/x)^8$.

    A. $48384x^3$ B. $48483x^3$ C. $48483x^{-3}$ D. $48384x^{-3}$

13. Obtain the sixth term in the expansion of $(2 + \frac{1}{2}x)^8$.

A. $14x^5$ B. $14x^6$ C. $28x^5$ D. $28x^6$

14. Obtain the fourth term in the expansion of $(x - y)^9$.

A. $84x^6y^3$ B. $-84x^6y^3$ C. $126x^5y^4$ D. $-126x^5y^4$

15. Obtain the fourth term in the expansion of $(x^2 + 3)^6$.

A. $540x^4$ B. $540x^6$ C. $1215x^4$ D. $1215x^6$

16. Obtain the sixth term in the expansion of $(3x - 2y)^9$.

A. $145152x^3y^6$ B. $-145152x^3y^6$ C. $326592x^4y^5$ D. $-326592x^4y^5$

17. Obtain the second term in the expansion of $(2x - 1)^2$.

A. 4 B. -4 C. $-4x$ D. $4x$

18. What is the coefficient of x^3 in $(2 + 3x)^4$

A. 54 B. 72 C. 81 D. 216

19. What is the coefficient of x^2 in$(1 + 2x)^{-2}$

A. 3 B. 6 C. 12 D. 18

20. What is the coefficient of x^3 in $(x - \frac{1}{2})^8$.

A. $-\frac{7}{2}$ B. $-\frac{7}{4}$ C. $\frac{7}{2}$ D. $\frac{7}{4}$

21. Find the coefficient of the fifth term in the expansion $(2 + x^2)^8$

A. $1120x^8$ B. $1120x^{10}$ C. $1792x^8$ D. $1792x^{10}$

22. Calculate a if the coefficient of x^3 in $(a + 2x)^5$ is 320.

A. 2 B. 4 C. 8 D. 16

23. Obtain the first three terms of the expansion $(1 + x)^{1/m}$.

A. $1 + \frac{1}{m}x + \frac{m-1}{2m^2}x^2$ B. $1 + \frac{1}{m}x - \frac{m-1}{2m^2}x^2$

C. $1 + \frac{1}{m}x + \frac{m+1}{2}x^2$ D. $1 + \frac{1}{m}x + \frac{m-1}{2m}x^2$

24. Obtain the first four terms of the expansion $(1 + 3x)^{1/3}$.

A.* $1 + x - x^2$ B. $1 + 3x - x^2$ C. $1 + 3x + 3x^2$ D. $1 + 3x - 3x^2$

25. Find the term independent of x in the expansion of $(x^2 + \frac{2}{x})^6$.

A. 96 B. 160 C. 240 D. 320

26. Find the term independent of y in the expansion of $(\frac{x^4}{y^3} + \frac{y^2}{2x})^{10}$.

A. $\frac{105}{16}x^2$ B. $\frac{105}{16}x^3$ C. $\frac{105}{32}x^2$ D. $\frac{105}{32}x^3$

27. Find the term independent of x in the expansion of $(\frac{x^4}{y^3} + \frac{y^2}{2x})^{10}$.

A. $\frac{45}{32}y^6$ B. $\frac{45}{64}y^8$ C. $\frac{45}{64}y^{10}$ D. $\frac{45}{256}y^{10}$

28. Find the term independent of x in the expansion of $(2x + \frac{3}{x^2})^6$.

A. 360 B. 720 C. 1080 D. 2160

29. Find the coefficient of x^5 in the expansion of $(x^2 + \frac{2y}{x})^{10}$.

A. $3360y^5$ B. $3360y^6$ C. $8064y^5$ D. $8064y^5$

30. If $(1 + 4x)^5 = 1 + Ax - Bx^2 + Cx^3$ ..., find the values of $A + B + C$.

A. 360 B. 500 C. 320 D. 640

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ANSWERS							
SET THEORY				MAPPING AND FUNCTION			
1.	D	21.	C	1.	D	21.	A
2.	C	22.	D	2.	C	22.	D
3.	C	23.	C	3.	B	23.	C
4.	D	24.	B	4.	B	24.	D
5.	C	25.	A	5.	C	25.	A
6.	A	26.	D	6.	A	26.	A
7.	B	27.	D	7.	B	27.	A
8.	C	28.	B	8.	B	28.	C
9.	C	29.	D	9.	A	29.	A
10.	D	30.	D	10.	C	30.	D
11.	B	31.	A	11.	D	31.	C
12.	D	32.	B	12.	B	32.	A
13.	A	33.	A	13.	D	33.	D
14.	C	34.	D	14.	A	34.	C
15.	C	35.	B	15.	C	35.	A
16.	D	36.	E	16.	B	36.	C
17.	A	37.	C	17.	C	37.	D
18.	A	38.	C	18.	D	38.	A
19.	C	39.	C	19.	A	39.	C
20.	B	40.	A	20.	C	40.	C
41.	C	46.	B	41.	C	46.	A
42.	C	47.	B	42.	C	47.	D
43.	C	48.	C	43.	A	48.	D
44.	B	48.	B	44.	C	49.	C
45.	C	50.	C	45.	D	50.	D

ANSWERS							
COMPLEX NUMBER				INEQUALITY			
1.	C	21.	D	1.	B	21.	D
2.	A	22.	A	2.	D	22.	B
3.	D	23.	C	3.	A	23.	A
4.	B	24.	B	4.	B	24.	C
5.	A	25.	A	5.	B	25.	A
6.	D	26.	C	6.	C	26.	B
7.	B	27.	D	7.	B	27.	D
8.	B	28.	A	8.	D	28.	D
9.	B	29.	A	9.	C	29.	A
10.	A	30.	C	10.	A	30.	C
11.	C	31.	A	11.	B	31.	B
12.	C	32.	A	12.	D	32.	D
13.	A	33.	B	13.	D	33.	B
14.	D	34.	B	14.	A	34.	B
15.	A	35.	D	15.	D	35.	C
16.	D	36.	D	16.	C	36.	C
17.	A	37.	A	17.	B	37.	C
18.	B	38.	D	18.	C	38.	C
19.	A	39.	C	19.	C	39.	A
20.	A	40.	C	20.	B	40.	B
41.	C	46.	D	41.	A	46.	B
42.	A	47.	C	42.	A	47.	C
43.	A	48.	D	43.	D	48.	C
44.	A	48.	C	44.	D	49.	C
45.	B	50.	C	45.	B	50.	A

ANSWERS							
SURD				INDICES AND LOGARITHMS			
1.	B	21.	D	1.	B	21.	B
2.	A	22.	B	2.	B	22.	C
3.	D	23.	B	3.	A	23.	B
4.	D	24.	A	4.	D	24.	B
5.	A	25.	D	5.	B	25.	C
6.	B	26.	D	6.	C	26.	C
7.	C	27.	B	7.	D	27.	D
8.	A	28.	C	8.	C	28.	C
9.	B	29.	D	9.	A	29.	B
10.	A	30.	A	10.	A	30.	D
11.	D	31.	A	11.	B	31.	D
12.	A	32.	C	12.	C	32.	D
13.	C	33.	B	13.	A	33.	C
14.	B	34.	C	14.	B	34.	A
15.	C	35.	D	15.	C	35.	C
16.	B	36.	B	16.	A	36.	A
17.	A	37.	D	17.	D	37.	B
18.	B	38.	B	18.	A	38.	B
19.	D	39.	B	19.	B	39.	C
20.	D	40.	D	20.	A	40.	D
41.	A	46.	D	41.	A	46.	C
42.	A	47.	B	42.	C	47.	A
43.	D	48.	D	43.	A	48.	A
44.	C	48.	C	44.	C	49.	D
45.	D	50.	B	45.	D	50.	C

BSG BRAINSTORM GROUP

ANSWERS					
MATHEMATICLA INDUCTION		BINOMIAL EXPANSION			
1.	A	1.	C	21.	A
2.	B	2.	C	22.	A
3.	B	3.	A	23.	B
4.	C	4.	D	24.	A
5.	D	5.	C	25.	C
6.	D	6.	D	26.	C
7.	B	7.	D	27.	D
8.	B	8.	B	28.	D
9.	B	9.	C	29.	C
10.	A	10.	C	30.	B
11.	D	11.	B		
12.	C	12.	D		
13.	D	13.	A		
14.	B	14.	A		
15.	A	15.	B		
16.	A	16.	D		
17.	C	17.	C		
18.	B	18.	D		
19.	C	19.	C		
20.	C	20.	D		